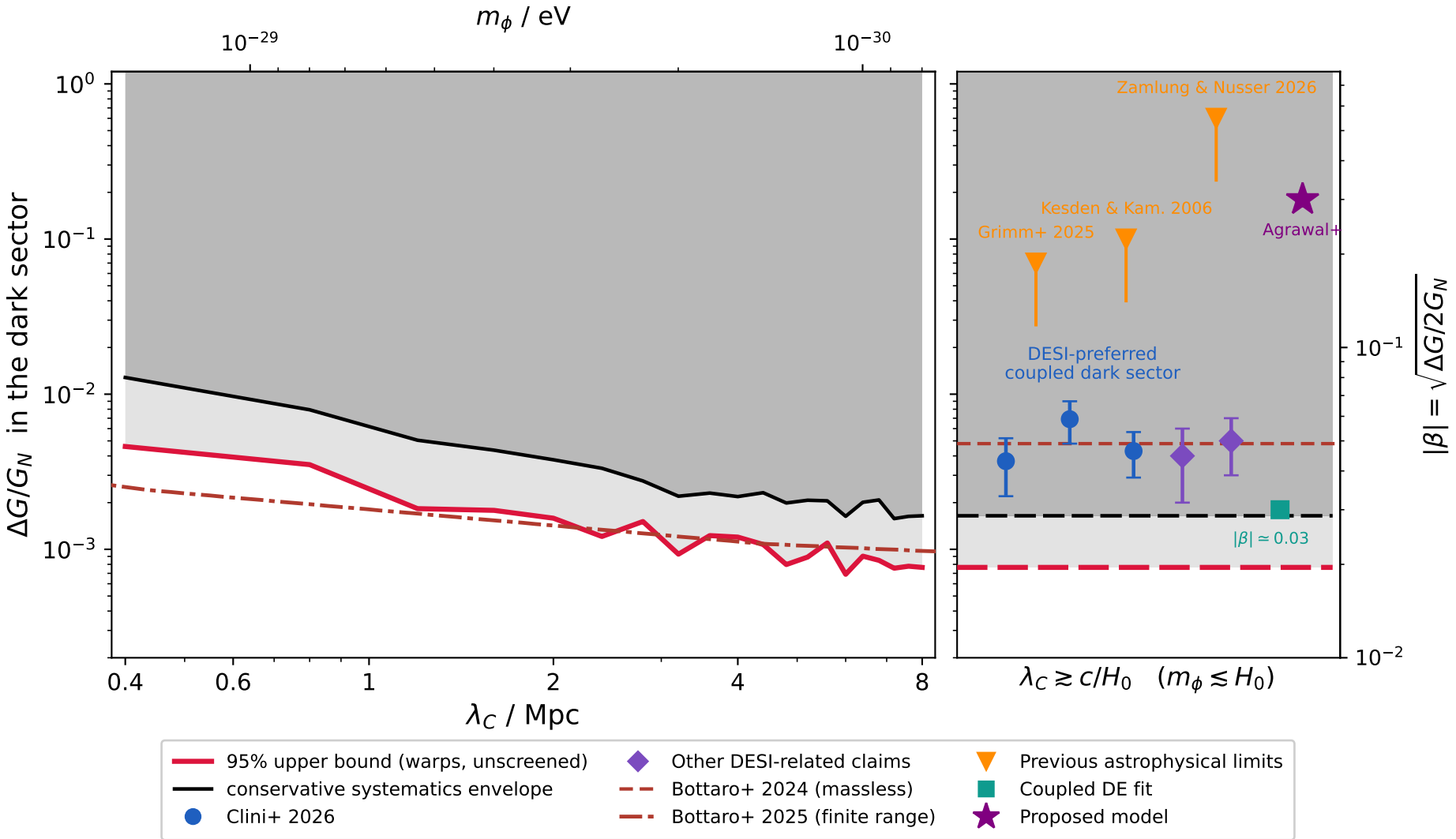
m_ϕ / eV
10^{-29}
10^{-30}
10^0
10^{-1}
10^{-2}
10^{-3}
$\Delta G/G_N$ in the dark sector
0.4
0.6
1
2
4
8
λ_C / Mpc
Zamlung & Nusser 2026
Kesden & Kam. 2006
Grimm+ 2025
Agrawal+
DESI-preferred
coupled dark sector
$|\beta| \simeq 0.03$
$\lambda_C \gtrsim c/H_0 \quad (m_\phi \lesssim H_0)$
10^{-1}
10^{-2}
$|\beta| = \sqrt{\Delta G/2G_N}$
95% upper bound (warps, unscreened)
conservative systematics envelope
Clini+ 2026
Other DESI-related claims
Bottaro+ 2024 (massless)
Bottaro+ 2025 (finite range)
Previous astrophysical limits
Coupled DE fit
Proposed model